Technology Stack

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs used to build OptiCrop. The stack below reflects the actual technologies and libraries used in the implemented project.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap-style responsive layout (custom style.css)	Lightweight, responsive UI for the Home, About, and Find Your Crop pages; no build tooling needed for a simple form-based app.
2	Backend / Server-Side	Python 3.10+, Flask	Flask is lightweight, integrates directly with scikit-learn/pickle, and is well-suited for quickly deploying an ML model behind simple routes (/, /about, /findyourcrop).
3	Machine Learning	Scikit-learn (KNN, Logistic Regression, Decision Tree, Random Forest, K-Means)	Provides a full suite of classification and clustering algorithms used to compare models and select the best-performing one (Random Forest) for crop prediction.
4	Data Processing & Storage	Pandas, NumPy, CSV dataset (crop_data.csv), Pickle (.pkl)	Pandas/NumPy handle data loading, cleaning, and feature engineering; the trained model and scaler are serialized with Pickle so they can be reused instantly at prediction time without retraining.
5	Visualization / EDA	Matplotlib, Seaborn (fivethirtyeight style)	Used during EDA (Epic 2) for univariate, bivariate, and multivariate analysis, plus the K-Means elbow plot and confusion matrix during model evaluation.
6	Development Environment	Anaconda Navigator, Jupyter Notebook, VS Code / PyCharm	Jupyter Notebook supports iterative EDA and model experimentation; VS Code/PyCharm is used for building and running the Flask application.
7	Version Control & Deployment	Git, GitHub; Flask development server (local)	Git/GitHub track project history and enable collaboration; the Flask dev server (`python app.py`) is used for local testing and demonstration.

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8	Third-Party APIs / Other Tools	None currently integrated (self-contained ML model)	The current version relies entirely on the trained model and static dataset; a future enhancement could add a weather API for real-time rainfall/temperature input.